CALLEY NYE

Design Engineer · Accessibility Specialist · Trustworthy AI Interfaces

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Full Sail University, Web Design & Development · 2010–2011

SUMMARY

Design engineer with 10+ years building consumer products, specializing in accessibility since my first engineering role in 2013. I own work from problem diagnosis through shipped interface: finding where a product's premise breaks, designing what people need in order to trust it, and building it in production code. Most recently, I re-scoped an AI recommendation product around what its data could honestly support, and built a Claude-driven design workflow that critiques every component for accessibility and overconfident AI messaging before it ships. **Interfaces designed for the hardest conditions end up better for everyone**, and AI-assisted development finally makes that the path of least resistance. I ship in React, Next.js, and TypeScript, with deep production experience in Vue 3, real-time UIs, and design systems.

EXPERIENCE

Founder & Design Engineer · Calley Co · Los Angeles *March 2023 — Present*

Independent design and engineering studio building AI-powered products, automation, and accessible interfaces for startups, agencies, and e-commerce brands.

- Re-scoped an **AI size-recommendation product** for a pre-launch school uniform brand whose local, in-person delivery made every return a costly pickup. Found the existing age-and-weight model had no measurement data behind it, and redesigned around what the data could honestly support: a measurement-first path, a faster estimate path labeled as less accurate, per-product size charts, and a branded measuring tape in every first delivery.
- Built and directed a **Claude design workflow** (three variants per component, automated critique for accessibility, brand voice, and overconfident AI messaging, then refinement) that shipped 9 recommender components. Product-managed an **AI size-chart manager** that parses supplier PDFs for human review, and specced the returns, exchange, and trade-in flows across customer, driver, and warehouse.
- Moving a marketing agency's senior living client sites off site-builder WordPress and an accessibility overlay plugin, onto **accessible-by-default static sites** built with Claude, versioned in GitHub, and deployed on Cloudflare, reducing clients' ADA lawsuit risk with projected savings of **$2,500+/month** in hosting and software.
- Build **Claude-driven automation workflows** for agencies and e-commerce brands across product development, merchandising, and content operations.
- **Accessibility and performance audits and remediation** for web products: WCAG 2.1/2.2 AA conformance, Core Web Vitals, and prioritized fixes teams can ship.

Senior Software Engineer · McKinsey & Company · Los Angeles *June 2021 — July 2023*

Game-Based Innovations Lab — game-based assessments and leadership training experiences used globally.

- Built a **distributed real-time multiplayer experience over WebSockets** for a COVID-era management leadership workshop — globally synchronized shared state, turn-based control passed between players, and a drag-and-drop interface where each player's interactions rendered live on every other player's screen.
- Designed and built the frontend for two assessment and training experiences in **Vue 3, TypeScript, and Node.js**, working directly with designers and product managers to translate detailed design intent into polished, accessible production code.
- Architected the team's **Vue component library and design system** — the shared foundation all current and future GBIL experiences are built on — with accessibility, performance, and reusability as core architectural requirements, not afterthoughts.
- Championed accessibility across the team through code reviews, formal discussions, and advocating for accessibility as a hiring-process equity issue — and led implementation of event analytics for product research.

- Contributed to Access McKinsey — the firm's disability affinity group — whose thought leadership reaches McKinsey's global workforce and the broader industry through its publishing channels.

Expert Frontend Developer (Contract) · Milliman · Remote *December 2020 — May 2021*

WCAG 2.0 conversion of SaaS interfaces — forms, data grids, dashboards — from ASP.NET Webforms to MVC.

- Delivered a **responsive, WCAG 2.0-compliant overhaul spanning four disparate codebases**, translating complex data-heavy designs into accessible production code.
- Designed and built a new **global navigation experience** unifying four previously disconnected products across ASP.NET MVC, Web Forms, and legacy SCSS systems.
- Proposed and implemented a modernized build process: watch mode, autoprefixing, transpilation, mobile device emulator testing, and the foundation of a shared UI library.

Founder & Principal Engineer · Syren.io · Los Angeles *November 2011 — November 2021*

Design, development, and product strategy consultancy applying lean product development and empathy-driven design to complex technical problems.

- Delivered frontend and full-stack solutions on **40+ client engagements**, often stepping into large legacy codebases to identify architectural debt, propose refactors, and ship measurable improvements.
- Drove growth strategy and developer relations for multiple fintech and crypto startups, including growing a community channel from **zero to 100K members**.

UI Engineer & Co-Founder · Popularium (acquired) · Los Angeles *February 2016 — December 2018*

Storytelling-driven product recommendation platform — real stories about real products from real people.

- Built the initial platform end-to-end using Django, designing user experience and core journeys alongside the engineering.
- Established a custom **testing and analytics framework** that informed product and editorial decisions.

UX Developer · Crowdrise (acquired by GoFundMe) · Los Angeles *September 2013 — February 2016*

Charity crowdfunding platform used by many of the world's largest nonprofits.

- Designed and shipped a higher-converting responsive signup process that lifted completion rates across the funnel.
- Led implementation of a new feature that became **20% of the company's business** within its first year of launch.
- Established frontend standards in SASS and JavaScript and integrated the Foundation framework to accelerate responsive development across the team.

SELECTED WRITING & SPEAKING

- **Which model of disability is your AI product operating from?** — Essay and interactive microsite mapping society's evolving models of disability onto patterns in modern AI interface design, and arguing AI-assisted development is the leverage point that makes inclusive engineering economically viable for the first time. calleynye.com
- **What is a Disability?** — Internal talk at McKinsey (Dec 2022) on the five models of disability and their implications for how we build recruiting products. Intellectual foundation for my current work on AI and accessibility.
- **JS.LA** (March 2015) — Conference talk on JavaScript and frontend practice. youtu.be/zKwNkX7MU3c · **Crypto Invest Summit** — Women of Crypto Track (2018), Marketing Workshop panelist · **DeveloperWeek 2015** speaker.

TECHNICAL SKILLS

Currently shipping · React, Next.js, TypeScript, Node.js, Vue 3 (Composition API), Tailwind CSS, Cloudflare, Claude Code and AI-assisted design and development workflows

Accessibility · WCAG 2.0 / 2.1 / 2.2 AA remediation, WAI-ARIA patterns, semantic HTML, focus management, keyboard navigation architecture, screen reader testing (NVDA, VoiceOver, JAWS), accessible streaming/live-region patterns for dynamic content

Real-time & streaming UIs · WebSockets, shared state synchronization, event-driven architecture, multiplayer interactions, accessible live-region patterns for streaming content, optimistic UI updates

Performance & Design Systems · Core Web Vitals optimization, Lighthouse auditing, bundle analysis, component library architecture, Storybook, design token systems, cross-codebase design system unification